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## ✓ Customer Preference Analysis

### ✓ Analyzing the relationship between the Type of Cuisine and the Restaurant's Rating

```
1 import pandas as pd
2 data = pd.read_csv('Dataset .csv')
```

```
1 data.columns
```

```
⇒ Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
        'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
        'Average Cost for two', 'Currency', 'Has Table booking',
        'Has Online delivery', 'Is delivering now', 'Switch to order menu',
        'Price range', 'Aggregate rating', 'Rating color', 'Rating text',
        'Votes'],
        dtype='object')
```

### ✓ Calculating the avg rating for each type of cuisine

```
1 avg_rating_cuisine = data.groupby('Cuisines')['Aggregate rating'].mean()
2 #grouping data by 'Cuisines' and then calculating the mean for 'aggregate rating' for eac
3 print('Average rating for each type of cuisine:')
4 print(avg_rating_cuisine)
```

```
⇒ Average rating for each type of cuisine:
Cuisines
Afghani                                0.725
Afghani, Mughlai, Chinese              0.000
Afghani, North Indian                 0.000
Afghani, North Indian, Pakistani, Arabian 0.000
African                               4.700
...
Western, Asian, Cafe                  4.200
Western, Fusion, Fast Food            3.200
World Cuisine                        4.900
World Cuisine, Mexican, Italian       4.400
World Cuisine, Patisserie, Cafe       4.200
Name: Aggregate rating, Length: 1825, dtype: float64
```

## ✓ Identifying most popular Cuisines based on votes

### ✓ Grouping the columns based on 'Cuisines' and calculating vote for each

```
1 total_votes = data.groupby('Cuisines')['Votes'].sum()
2 print('Total votes for each cuisine:')
3 print(total_votes)
```

```
→ Total votes for each cuisine:
Cuisines
Afghani                    39
Afghani, Mughlai, Chinese    2
Afghani, North Indian        0
Afghani, North Indian, Pakistani, Arabian    3
African                    373
...
Western, Asian, Cafe        259
Western, Fusion, Fast Food   32
World Cuisine               95
World Cuisine, Mexican, Italian    115
World Cuisine, Patisserie, Cafe    1034
Name: Votes, Length: 1825, dtype: int64
```

### ✓ Identifying the most popular cuisines by sorting

```
1 most_popular_cuisines = total_votes.sort_values(ascending=False).head(10)
2 print('Most popular cuisines:')
3 print(most_popular_cuisines)
```

```
→ Most popular cuisines:
Cuisines
North Indian, Mughlai        53747
North Indian                46241
North Indian, Chinese        42012
Cafe                        30657
Chinese                     21925
North Indian, Mughlai, Chinese    20115
Fast Food                   17852
South Indian                16433
Mughlai, North Indian        15275
Italian                    14799
Name: Votes, dtype: int64
```

From this, we can understand that cuisines like North Indian - Mughlai, Chinese, Cafe, Fast food etc have received the highest numbers of votes and are the most popular cuisines

## ✓ Determining if there are any specific cuisines that tend to receive higher ratings

```
1 topRatedCuisines = avg_rating_cuisine.sort_values(ascending=False)
2 print('Top rated cuisines:')
3 print(topRatedCuisines)
```

```
→ Top rated cuisines:
Cuisines
Italian, Deli          4.9
Hawaiian, Seafood     4.9
American, Sandwich, Tea 4.9
Continental, Indian   4.9
European, Asian, Indian 4.9
...
Fast Food, Desserts, Beverages 0.0
Fast Food, Chinese, Burger 0.0
Cafe, French            0.0
Seafood, Mughlai, North Indian 0.0
Fast Food, Mughlai, Armenian 0.0
Name: Aggregate rating, Length: 1825, dtype: float64
```

The identified cuisines with high ratings and most votes reflect specific customer preferences, showcasing popularity.

